Programming Fundamentals
Lab-03

Task 1:

Write a program to print the table of a number, with the following ways:

a) Using only cout statement
b) Using a variable
c) Taking input from the user the number of which the table user wants
d) Using while loop
e) Asking user the start and end of the table

Task 2:

a) Write a program to print an odd number table using a while loop
b) Write a program to print an even number table using a while loop

```
----------Odd table Printing----------
Enter the number of which you want table: 16
Enter the range: 10
   16x1=16
   16x3=48
   16x5=80
   16x7=112
   16x9=144
```

Task 3:

Write a program that asks the start and end from the user and then prints the table in reverse order.

```
Enter the number of which you want table: 15
Enter the start: 10
Enter the end: 5
15 x 10 = 150
15 x  9 = 135
15 x  8 = 120
15 x  7 = 105
15 x  6 = 90
15 x  5 = 75
```

Task 4:

a) Write a program that ask for the user a value of jump and then prints the table of a number according to that jump
b) Now make a reverse version of it.

```
Enter the number of which you want table: 10
Enter the start: 10
Enter the end: 25
Enter the value of jump: 5
10 x 10 = 100
10 x 15 = 150
10 x 20 = 200
10 x 25 = 250
```

Task 5:

a) Write a program to print cubes of number given as input

b) Write a program that takes starting and ending point as input and print the cubes of numbers within the range

```
Ente the number: 8
The cube of 8 is: 512
```

```
Enter the starting number: 2
Enter the ending number: 9
The cube of 2 is: 8
The cube of 3 is: 27
The cube of 4 is: 64
The cube of 5 is: 125
The cube of 6 is: 216
The cube of 7 is: 343
The cube of 8 is: 512
The cube of 9 is: 729
```

Task 6: Sequence Printings

Write programs that generate the following patterns:

a) 1 3 5 7 9 11 … 99 (All odd numbers less than 100)

b) 0 2 4 6 8 10 … 98 (All evens lesser than 100)

c) 99 97 95 . . . 7 5 3 1 (All odd numbers less than 100, in reverse order)

d) 98 96 94 92 … 8 6 4 2 0 (All evens lesser than 100, in reverse order)

e) 100 50 25 12 6 3 1 (keep decreasing by division of 2 until the remainder is >1)

Task 7: Temperature Conversions

A. Write a program that prompts the user to enter a temperature in Celsius (C) from the keyboard. The program should then convert the temperature to both Fahrenheit (F) and Kelvin (K) using the following formulas:

Fahrenheit ***(F) = (9/5 * C) + 32***

Kelvin (K) = ***C + 273.15***

Sample Input: C = 12.3

Sample Output: F = 54.14, K = 285.

B. Write a program that prompts the user to enter a Fahrenheit (F) temperature. The program should then convert the temperature to both Celsius (C) and Kelvin (K) using the following formulas:

Celsius (C) = ***(F - 32) * 5/9***

Kelvin (K) = ***(F - 32) * 5/9 + 273.15***

Sample Input: F = 54.14

Sample Output: C = 12.30, K = 285.45

C. Write a program that prompts the user to enter a temperature in Kelvin (K). The program should then convert the temperature to both Celsius (C) and Fahrenheit (F) using the following formulas:

Celsius (C) = ***K - 273.15***

Fahrenheit (F) = ***(K - 273.15) * 9/5 + 32***

Sample Input: K = 285.45

Sample Output: C = 12.30, F = 54.14

Task 8: N^K

Write a program that takes two numbers, K and N, and calculates N raised to the power of K (N^K). Finally, display the result.

Sample Input:

K = 3

N = 2

Sample output:

8

Task 9: Swapping Integers

Write a program that takes two integers A and B from the user and swaps their values using a third temporary variable.

Sample Output: Before Swapping: **A = 45** **B = 32**

After Swapping: **A = 32** **B = 45**

BONUS: To solve the same problem, swap the values of two integers, **A** and **B**. **You are not allowed to use any extra variables other than A and B.**

Task 10: Average of K numbers

Take K integers from the keyboard using a loop and print their average value on the screen.

Sample Input:

K=10

integers: 2 5 6 14 10 17 12 7 1 3

Sample Output: **Average Value: 7.7**

Task 11: Sum of K numbers

Write a program that first asks the user a number **K**. Then take **K** integers from the keyboard using a loop and print their sum on the screen.

Sample Input: K = 6

3 12 7 2 54 32

Sample Output: **110**

Task 12: Printing multiples less than N

A. Write a program that prompts the user to enter a positive integer (N) and then prints the sum of all the multiples of 3 or 5 that are less than N

Sample Input:

N = 20

Sample Output:

The sum of multiples of 3 or 5 less than 20: 78

Task 13: Factorial

Write a program to calculate the factorial of a number.

How to calculate the factorial of a number?

To calculate the factorial of N you just have to multiply all the numbers between N and 1.

For Example:

Factorial of N: N x (N - 1) x (N - 2) x (N - 3) ……1

Sample Input:

Number: 5

Sample Output:

Factorial of 5 is 120

Task 14: Gym Tracker

Taha owns a gym. Taha wants to know how many times a person can perform a particular exercise before reaching muscle fatigue. The person starts with a certain number of repetitions and decreases the number of repetitions by a fixed amount after each set until they can no longer perform the exercise.

Sample Input:

startReps = 10
decrement = 2
minReps = 4

Sample Output:

Set 1: 10 reps
Set 2: 8 reps
Set 3: 6 reps
Set 4: 4 reps
Total sets completed: 4

Explanation:

- The person starts by doing 10 repetitions.
- In the next set, they decrease by 2 reps, so they do 8.
- This continues until they reach the minimum number of repetitions (4 in this case).
- The program outputs each set's repetitions and the total number of sets.

Task 15: Parking lot

You are creating a program for a parking lot that charges fees based on the number of hours a vehicle is parked. The parking fee is calculated on an hourly basis, but the rates change based on the time spent in the lot:

- The first hour costs $5.
- Each additional hour up to 5 hours costs $3.
- After 5 hours, each additional hour costs $2.

The program should calculate the total parking fee for a vehicle based on the number of hours it was parked.

Sample Input:

Hours = 7

Sample Output:

Total parking fee: $21

Explanation:

- For 7 hours:
  - The first hour costs $5.
  - The next 4 hours (up to 5 hours total) cost $3 each, totaling $12.
  - usThe remaining 2 hours (beyond 5 hours) cost $2 each, totaling $4.
  - Total fee: $5 + $12 + $4 = $21.

HINT: More than one individual loop is used in this question.

Task 16: Interest System

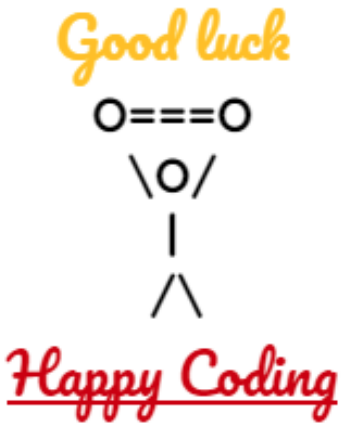
You are developing a simple financial application that calculates the total amount in a savings account after a certain number of years. The initial amount in the account is N, and it earns a fixed interest of I per year. Write a program that calculates the amount in the account after Y years.

Sample Input:

Initial Amount: $1000

Interest Rate: 5 //This is in percentage

Number of years: 10

Sample Output:

After 10 years the amount increase to $1628.91